PROGRESS REPORT

Master of Applied IT

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HBO-i Architectural Layers

Level 4 Domain: Organisational Processes
Level 3 Domain: Software Engineering

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1. Project Overview

The project investigates how semi-supervised learning (SSL) and synthetic dermatological images can contribute to cost-efficient, sustainable, and fair AI solutions for skin cancer detection.

The central research question is:

How can semi-supervised learning and synthetic data help achieve cost-efficient, sustainable AI software solutions for skin-cancer detection?

This is broken down into three sub-questions around:

1. Data generation and labelling
2. SSL workflow design
3. Validation and cost-performance evaluation

The work is embedded in the Fontys Research Group “Sustainable Data & AI Applications”, with a strong focus on fairness (skin-type and sex bias), privacy, and cost-efficient annotation.

2. Progress on Project Deliverables

This section reflects the progress against the deliverables outlined in the Project Plan and Study Plan, and how these are visible in the portfolio.

2.1 Project Plan (Completed)

Planned deliverable:
A research project plan describing context, research gap, research questions, methods, deliverables, stakeholders, and planning.

Evidence in portfolio:

- “Project Plan - Synthetic Data Generation for Skin Cancer Detection.pdf”

Main contributions / value so far:

- Clearly articulates the **problem statement** (data scarcity, privacy, and equity issues in dermatological AI).
- Defines the **research gap** around integrating SSL with synthetic data for bias mitigation, cost-efficiency, and clinical validation.
- Specifies a **mixed-methods research design** (literature study, EDA, expert interviews, prototyping, multi-criteria decision making, validation and benchmark testing).
- Provides a **16-week timeline** and milestones (literature review by week 3, PoC by week 8, validation framework by week 10, experiments by week 12, final reporting by week 16).

This plan has been used as the main reference to steer the work and to align expectations with the supervisor and domain expert.

2.2 Background Research – Literature Review (Completed)

Planned deliverable:
An extensive literature review / background research report on dermatology datasets, bias, synthetic data generation and SSL.

Evidence in portfolio:

- “Background research - Literature Review”

Content and value:

- Summarises the **clinical stakes** of skin-cancer detection and the limitations of major datasets (ISIC, HAM10000, BCN20000, Fitzpatrick17K), including representation gaps across Fitzpatrick skin types and sex.

- Surveys **generative approaches** (e.g. conditional GANs, diffusion models) and their implications for fairness, domain shift, and privacy.
- Describes **SSL algorithms** (such as FixMatch, MixMatch), integration of synthetic data, and fairness-aware pseudo-label filtering.
- Outlines **validation metrics and protocols** (ROC-AUC, PR-AUC, sensitivity at fixed specificity, calibration curves, stratified test sets, external validation), and discusses **risks** such as synthetic-to-real domain shift and generalisation across datasets.

This deliverable directly supports the **Organisational Processes – Analysis** learning outcome (framing a complex context and surfacing generalisable problems such as bias and labelling cost).

2.3 Proof-of-Concept Streamlit Application (Completed – Version 1)

Planned deliverable (Project Plan / Study Plan):
An end-to-end pipeline proof-of-concept for synthetic data generation, (semi-)automatic labelling, and model evaluation. Version 1 of the PoC is implemented as a **Kedro-based MLOps application with a Streamlit web UI**, focusing currently on synthetic image generation, integration with HAM10000, classification, and a training pipeline skeleton.

Evidence in portfolio:

- Source code repository (Kedro pipelines, model utilities, configuration).
- Streamlit web interface implementing the main workflows (Run Pipeline, Results, Saved Models).
- Internal documentation / README.

Technical scope implemented:

1. **Image Generation Pipeline**
    - Integrates multiple **diffusion and image models**:
        - Stable Diffusion v1.5
        - Stable Diffusion 3.5 Large
        - Qwen-Image
        - FLUX.1-dev

   - Supports **text-to-image** and **image-to-image** generation with HAM10000 reference images.
   - Standardises output to **600×450** to match HAM10000.
   - Includes batch generation and GPU/CPU memory management (CUDA cache clearing, CPU offloading, PyTorch compatibility patches).

2. **Prompt Engineering System**
   - Diagnosis-specific prompts for the 7 HAM10000 classes:
     - akiec (Actinic keratoses)
     - bcc (Basal cell carcinoma)
     - bkl (Benign keratosis)
     - df (Dermatofibroma)
     - nv (Melanocytic nevi)
     - mel (Melanoma)
     - vasc (Vascular lesions)
   - ISIC-style dermoscopic descriptions, clinical terminology, and negative prompts that exclude non-lesion content (e.g. body parts).
   - Generation modes for **single diagnosis** and **mixed distributions** with percentage control.

3. **HAM10000 Dataset Integration**
   - Loader that combines images and metadata (age, sex, localisation, diagnosis).
   - Filters by diagnosis/localisation/demographics and provides summary statistics.
   - Image-guided generation with configurable reference strength and reference percentage.

4. **Classification Pipeline**
   - Integration of pre-trained models (VGG-16, Inception v3, ResNet50) from the ENHANCE repository.
   - Batch classification, confidence scores, and class distribution analysis.
   - Result aggregation and summary statistics.

5. **Model Training Pipeline (Skeleton)**
    - Configurable data splits (e.g. 70/15/15) with **synthetic + real data mixing**.
    - Hyperparameter configuration (epochs, learning rate, batch size).
    - Model versioning and metadata storage hooks prepared.
6. **Web Interface (Streamlit)**
    - Tabs: **Run Pipeline**, **Results**, **Saved Models**.
    - User-configurable parameters for:
        - Generation model, number of images, diagnosis distribution, reference settings.
        - Training parameters and classification model selection.
    - Real-time progress reporting, image gallery, statistics, and debug information.
7. **Kedro-based Architecture**
    - Modular pipelines for generation, classification, and training.
    - Data catalog and parameter configuration.
    - Clear separation between data, pipeline logic, and UI.

Delivered value so far:

- Provides an **operational end-to-end pipeline** from **prompt + HAM10000 metadata → synthetic images → classification → optional training**, which directly instantiates the architecture defined in the planning.
- Enables systematic experimentation with **different generation models and synthetic/real mixes**, a prerequisite for answering the research sub-questions about cost-efficiency and performance.
- Serves as concrete evidence for the **“Realise”** learning outcomes in both Organisational Processes and Software domains for this semester.

At this stage, the PoC focuses on deterministic pipelines and manual analysis of results. The more advanced SSL loop (e.g. FixMatch, pseudo-label filtering) and full cost/benefit evaluation are planned for the second half of the project.

3. Progress on Learning Goals

The Study Plan sets the target of **Level 4 in Organisational Processes** and **Level 3 in Software** (Analysis, Advise, Design, Realise, Manage & Control), plus Professional, Personal, and Academic Standards.

Below is how current progress maps to these goals.

3.1 Organisational Processes – Level 4

Analysis:

- Completed a substantial **literature review** covering datasets, bias, synthetic data, SSL, and MLOps considerations.
- Project Plan and Study Plan together clearly articulate the **problem space, research gap, and context** (bias, privacy, annotation cost).

Advise:

- The literature review already positions different technical options (GAN vs diffusion, SSL variants, fairness interventions) with pros/cons.
- Initial choices for generation models and architecture in the PoC are grounded in this review; a more explicit decision matrix will be developed when comparing SSL strategies and backbone models.

Design:

- The **Kedro pipeline + Streamlit front-end** already reflect a modular, generalizable architecture that can be extended beyond HAM10000 or even dermatology.
- The target of an end-to-end architecture has been translated into concrete pipeline components and parameterisable configurations.

Realise:

Version 1 of the pipeline (generation → classification → training skeleton) is implemented and operational, matching the PoC milestone in the planning.

Manage & Control:

- The project already employs:
    - Structured **project plan and timeline**.
    - Reproducible configuration via Kedro (parameters, catalogs).
- In the coming phase, this will be strengthened with:

- Locked train/val/test splits.
- Standard metric set and benchmark protocol.
- Basic cost/annotation tracking, as planned in the Study Plan.

Overall, for Organisational Processes, the project has moved from an orienting phase to a clear **Beginning / lower-Proficient** level for several sub-outcomes, especially Analysis, Design, and Realise, supported by tangible artefacts.

3.2 Software – Level 3

Analysis:

- Requirements for the PoC (models, image formats, HAM10000 integration, batch handling, memory management) have been identified and implemented in code.
- Acceptance criteria are still partly informal (e.g. stability, visual plausibility, basic classification sanity checks); formal metric thresholds will follow with the validation framework.

Advise:

- The choice to use **pre-trained diffusion models** and existing classification backbones aligns with the project's goals (re-use of large models, focus on fairness and sample quality instead of training from scratch).
- Decisions are currently captured implicitly (in documents and code); a more explicit decision log is a clear improvement point.

Design:

- Designed and implemented:
    - Data loaders with metadata (age, sex, localisation, diagnosis) and filters.
    - Prompt templates for each of the 7 HAM10000 classes.
    - Pipelines for generation, classification, and training.
- Next steps: ratio scheduler for synthetic/real mixing and explicit quality-filter modules based on classifier or heuristic QC.

Realise:

- Core modules and a reproducible repo exist; the app is deployable as a Streamlit front-end on top of Kedro.
- Basic experiments and test runs have been executed, though automated testing and full experiment logging are still in an early stage.

Manage & Control:

- Versioning is present at the level of models and configuration in the PoC.
- Formal CI, structured experiment logging, and semantic versioning are planned but not fully in place yet.

Overall, the project is **on track for Software Level 3**, with the PoC as strong evidence for the Realise outcome, and clear next steps around testing, logging, and explicit decision documentation.

3.3 Professional Skills, Personal Leadership, Academic Standard

Professional Standard:

- Fairness, privacy (GDPR), and bias reporting, especially in the literature review are explicitly considered in the project design (e.g. stratified validation, synthetic-data provenance, domain shift analysis).

Personal Leadership:

- A clear development goal has been formulated (Org. Processes L4, Software L3), and the Study Plan and self-assessments are used to track progress.
- There has been active engagement with supervisors and domain experts, and their input has been translated into concrete artefacts (Project Plan, Literature Review, PoC).

Academic Standard:

- The literature review uses a broad and current academic base and is linked to methodological choices in the Project Plan and implementation in the PoC.
- The project combines theoretical knowledge (GANs vs diffusion, SSL algorithms, fairness metrics) with practical prototyping, showing investigative ability.

4. Planning for the Next Phase

The next half of the project will mainly cover **Phase 3 (Experimental Implementation and Validation)** and **Phase 4 (Documentation and Reporting)** as planned in both the Study Plan and the Project Plan.

Planned focus areas:

1. **Complete the SSL pipeline**
    - Implement SSL training on top of the existing pipeline (e.g. FixMatch/MixMatch variant).
    - Integrate fairness-aware pseudo-label filtering.
2. **Implement the validation and monitoring framework**
    - Locked train/val/test splits (real-only test sets).
    - Standard metrics (ROC-AUC, PR-AUC, sensitivity at 90% specificity, calibration error), including stratified analyses (by class and relevant demographics where available).
    - Domain-gap analyses (real vs synthetic vs hybrid).
3. **Run systematic experiments**
    - Compare models trained on real-only, synthetic-only, and hybrid datasets with varying ratios.
    - Measure annotation effort and cost proxies to assess cost-efficiency.
4. **Document architecture and decisions**
    - Finalise IT architecture sketch (data flow, training, deployment, monitoring, governance).
    - Create a multi-criteria decision matrix with transparent trade-offs.
5. **Final reporting**
    - Draft and refine the final research paper.
    - Produce technical documentation, a research poster, and handover materials.

5. Conclusion

By midterm, the project is **on track with the key early deliverables**:

- Project Plan – completed.
- Extensive Background Research / Literature Review – completed.
- A functional POC **Streamlit + Kedro** application that realises an end-to-end synthetic skin-lesion generation and classification pipeline – completed (version 1).
- Initial progress towards the Master learning outcomes in **Organisational Processes (L4)** and **Software (L3)**, with tangible evidence in the portfolio.

The remaining work will deepen the SSL component, formalise validation and cost-efficiency analysis, and translate the technical results into rigorous academic and practical outputs.